Experimental Diagnostics of Lightweight Countermeasures

Purpose. This document provides focused empirical diagnostics for the cyclic-shift and coded-index countermeasures in Section 8 of *Mind Your Oblivious Transfer (OT) Extensions Deployed in the Real-world*. The paper already states the constructions, primary numerical findings, and limitations. The four plots here expose additional *distributional* and *budget-dependent* evidence rather than new claims or an alternative presentation of the main results.
Evaluation context. The experiments use profiled, byte-level ranking. N counts attack examples; GE is the expected rank of the correct candidate and SR_k is its top-k success probability. Cyclic shift varies the 8-bit effective-delta relation Δ_{eff} and evaluates all 256 tested base-delta values. The protected settings use separately profiled attacker models. *Mechanism 4* refers to amplification through known adjacent-pair XOR offsets. These are empirical results for the tested configurations.
Evaluation settings. As recorded in the accompanying machine-learning experimental configurations, the protected ARM runs use two-channel differential inputs over samples 520–579 and a freshly trained protected model. The cyclic-shift sweep covers 256 base-delta values, up to 5,000 attack examples per value, and 100 randomized attack subsets per budget. The coded-index experiment uses a separate protected attack set, up to 10,000 attack examples, and 170 randomized subsets per budget.

1. Cyclic shift: offset-specific success

The cyclic-shift construction changes the effective relation between neighboring pairs by mixing a counter and its one-bit right rotation into the low byte of the target delta. The first diagnostic question is whether this modification uniformly prevents successful recovery, or instead concentrates success in a small set of effective-delta values. Figure 1 displays the observed rank-1 success rate across the tested 8-bit sweep.
Interpretation. The sparse peaks in this sweep support a narrower conclusion than universal protection: favorable effective-delta classes still exist, but strong success is not systematic across the tested values. This distinction motivates examining the *distribution of final ranks*, not just the maximum success rate.

2. Cyclic shift: distribution of final candidate ranks

Figure 2 complements the per-value success-rate sweep with two views of the final rank distribution over the 256 tested base-delta values. The cumulative curve and histogram summarize the range of attacker outcomes rather than a single selected favorable setting.

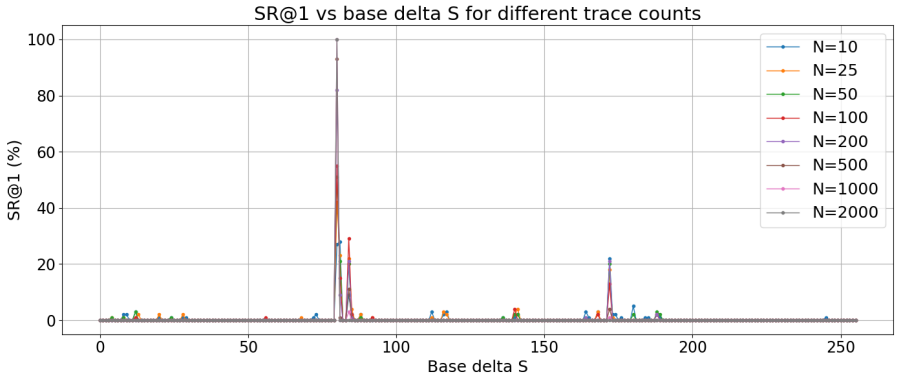


Fig. 1: SR_1 over the 8-bit effective-delta sweep under the cyclic-shift countermeasure. Only a few induced Δ_{eff} values yield visible SR_1 spikes, while most remain near zero. This indicates that strong few-trace success is concentrated in sparse effective-delta outliers rather than appearing systematically across the tested space.

Interpretation. In this experiment only one of the 256 tested base-delta values attained final rank 1. This is evidence about the tested 8-bit parameter space and evaluation procedure, not an assertion that all deployments have a universal success probability of 1/256. The original paper reports this observation in its countermeasure discussion; the distribution adds detail about the remaining rank values.

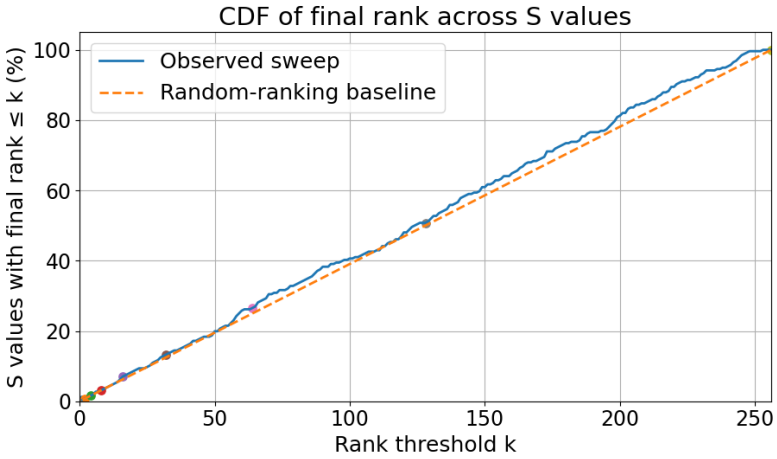
3. Comparison across protection settings

The second evaluation question is how the attack's ranking changes with the number N of attack examples when no countermeasure, cyclic shift, or coded indexing is applied. Figure 3 shows both GE and rank-1 SR against the same horizontal budget variable, making the trend and rate of convergence visible rather than reducing each curve to one endpoint.

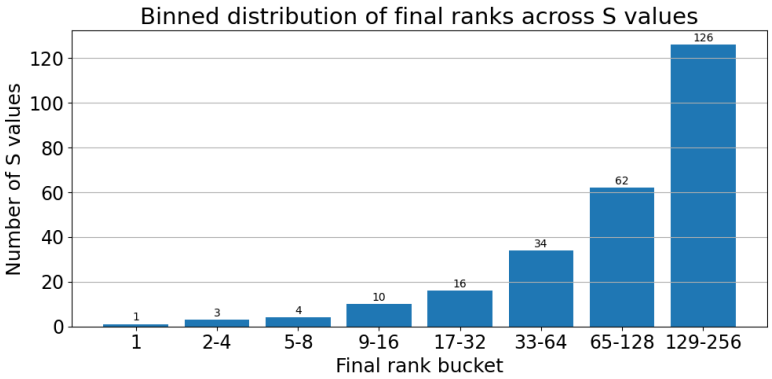
Interpretation. The unprotected setting converges quickly in the reported experiments. The protected settings degrade the observed attack performance, but the cyclic-shift sweep also shows that protection is not uniform over all effective-delta configurations. These curves support a comparison *within the evaluated implementation and attack settings*; they do not establish a general security ordering against all possible adversaries.

4. Coded index: behavior at a larger attack budget

Coded indexing replaces the adjacent 3-bit counter sequence with a coded sequence whose consecutive values have Hamming distance at least two. The remaining question is whether the profiled attacker converges as more examples

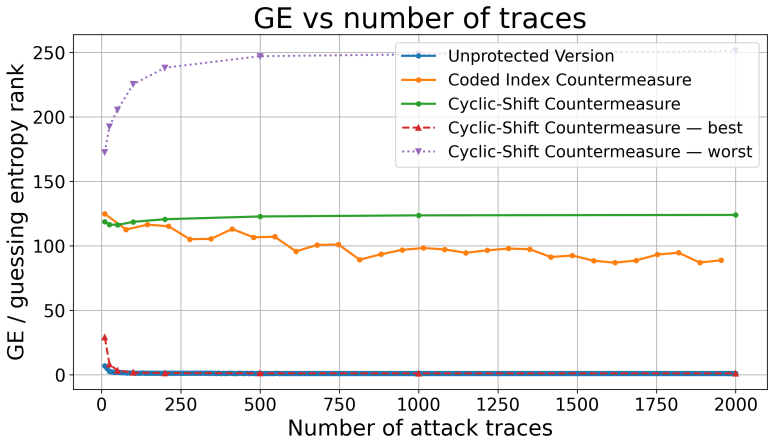


(a) Cumulative distribution

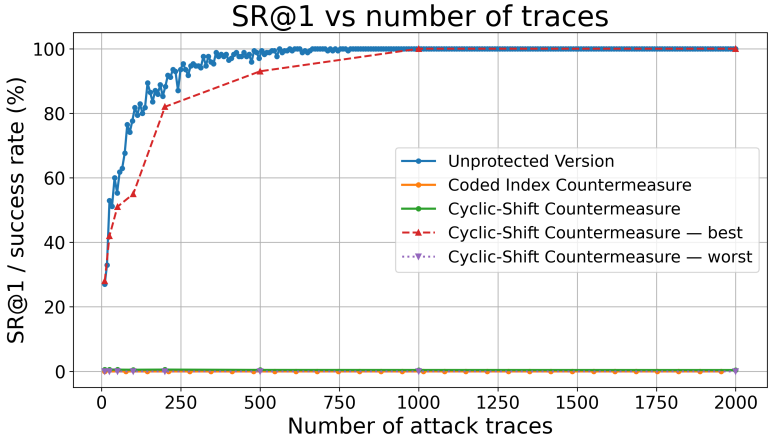


(b) Histogram.

Fig. 2: Distribution of final attack ranks under the cyclic-shift countermeasure across all 256 base-delta values (a) cumulative distribution (b) histogram.



(a)

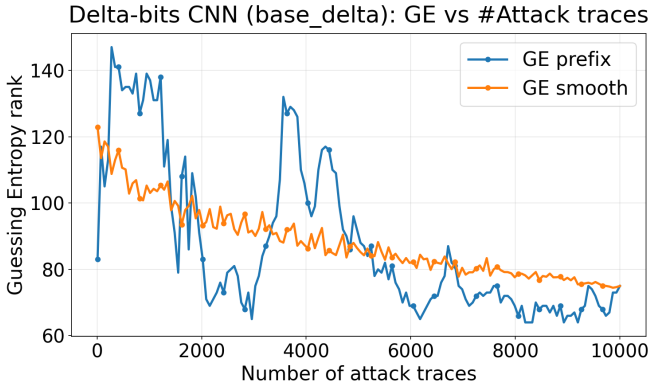


(b)

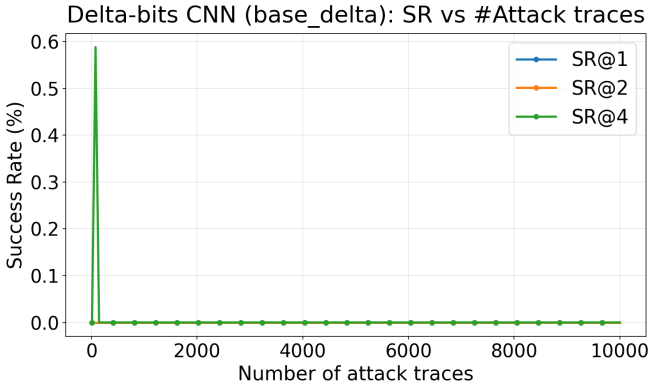
Fig. 3: Comparison of attack performance across countermeasure settings using (a) GE and (b) rank-1 SR as functions of N.

are aggregated. Figure 4 extends the view to the tested budget of 10,000 attack examples.

Interpretation and scope. The observed GE remains around rank 60 after 10,000 attack examples, while SR_1, SR_2, and SR_4 remain zero throughout this evaluation. These measurements indicate substantial degradation of the tested neighboring-pair attack path.



(a) Guessing entropy vs. attack examples.



(b) Success rate vs. attack examples.

Fig. 4: Attack performance under the coded-index countermeasure. The guessing entropy decreases only slowly and remains far from convergence even after 10,000 attack examples. In parallel, SR_1, SR_2, and SR_4 remain zero throughout the tested attack-example budget. This indicates that removing the low-Hamming-distance adjacency of *Mechanism 4* strongly degrades neighboring-pair amplification.